DNA Replication 2

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Telomerase (ribonucleoprotein) is an enzyme that has an active site that allows it to function as a “reverse transcriptase”, meaning it can synthesize DNA from an RNA template and contains an RNA template. It interacts with the end of a chromosome and adds repeated segments of noncoding DNA, a replisome then synthesizes the complementary strand.

Muscle, brain and liver cells lack active telomerase

Cancer originates in cells that have undergone replicative senescence (meaning that they no longer divide)

Repairing Replication Errors

DNA polymerase has exonuclease activity

- If it adds the wrong base (like G to A), then exonuclease will back up, remove the incorrect base from the 3’ end and add the correct base -> process is called proofreading
    - Exonuclease is an enzyme that can remove a nucleotide from the end of a nucleic acid strand

After replication is complete, specialized proteins scan the new double helix for mismatched bases. Once recognized, specialized enzymes cut around the mismatch and replace it with the correct deoxyribonucleotides.

When UV light hits DNA, it causes adjacent C’s and T’s to form covalent bonds with each other. The new bulge in the sugar-phosphate backbone prevents genes from being copied/transcribed normally.

To correct this damage, enzymes and protein:

1. Recognize the site of damage
2. Break the sugar-phosphate backbone on either side of the problem, open the double helix and remove the bases on the damaged strand
3. Replace the excised bases with the correct deoxyribonucleotides
4. Catalyze formation of phosphodiester linkages between the next section of bases and the adjacent DNA (reestablishes the backbone)